

Project Report

E-commerce Sales Analytics

1. Introduction

This project presents an end-to-end data analytics solution using the **Olist Brazilian E-commerce Dataset**, which contains information on customers, orders, products, payments, reviews, sellers, and deliveries. The analysis was conducted using **Python** for data cleaning and feature engineering, **SQL Server** for data storage and business analysis, and **Power BI** for developing interactive dashboards and visualizing key business metrics.

The primary objective of this project is to transform raw transactional data into meaningful business insights. Through structured analysis, the project identifies sales patterns, customer behavior, product category performance, delivery efficiency, and customer satisfaction. The findings aim to provide actionable recommendations that can help improve revenue, operational efficiency, and overall customer experience.

This report documents the complete workflow of the project, including data preparation, database design, SQL-based business analysis, dashboard development, and the key insights derived from the analysis.

2. Dataset Summary

The **Brazilian E-Commerce Public Dataset by Olist** contains transactional data from a Brazilian e-commerce marketplace, covering orders placed between **September 2016 and October 2018**. It includes approximately **100,000 orders** and consists of **8 interconnected tables** containing information on customers, orders, products, sellers, payments, reviews.

The dataset follows a relational database structure, allowing data from multiple tables to be linked using common keys such as **order_id**, **customer_id**, **product_id**, and **seller_id**. This enables comprehensive analysis of sales performance, customer behavior, product performance, payment methods, delivery efficiency, and customer satisfaction.

For this project, the dataset was cleaned and transformed using **Python**, analyzed using **SQL Server**, and visualized through interactive dashboards in **Power BI** to generate meaningful business insights and support data-driven decision-making.

3. Exploratory Data analysis using Python

1. Exploring the shape of all the tables
2. Using `df.info()` to find the data type of each column
3. Using `df.describe()` to view the statistical properties of numeric columns

4. Cleaning the Data Using Python

a) **Renaming column names**

using loop to convert name into lower case and replacing space with “_” underscore

b) **Removing The unnecessary columns**

removing columns like “zip_codes_prefix” from table “customers” and “sellers” .
“review_comment_title”, “review_comment_message”, “review_creation_date”,
“review_answer_timestamp” from review tables

c) **Removing The WhiteSpaces**

using Multi loops to remove the whitespaces in values

d) **Fixing the data type**

changing the data type from string to datetime in date columns of table “order_items” and “orders”

e) **Handling Nulls and Missing Values**

There are null values in order dataframe “order_df” . But Removing these null values rows is not a good idea as these are valid rows , if customer cancels the order then there the value in “order_delivered_customer_date ” will be null.

In product dataframe “product_df” have null values in column like
“product_category_name” and other dimension column,
replacing the null values in product_category_name to “unknown” and setting other column null to zero

f) **Removing Duplicate Rows**

Using loop to remove the to duplicate rows of each dataframes

g) **Reindexing**

5. Adding and Transforming the Columns

Creating New columns in order table named “order_df” , these columns are based on the date difference ‘the customer order timestamp’ with other stage completing timestamp ,and also the differences are converted in days.

New column are [“purchase_year”, “purchase_month”] which is calculated from column “order_purchase_timestamp” in orders tables.

“delivered_on_time” column which have only 2 values “Yes” and “No” , tells whether that order delivered on time .

Removing the unwanted columns from “order_df” like-

[‘order_delivered_carrier_date’, ‘order_delivered_customer_date’, ‘order_estimated_delivery_date’]

Mapping the product category name in “product_df” to English translation from “product_category_name_df” which carries the English translation of product category name.

6. Setting Connection with Python to SQL Server

After the Data cleaning and feature engineering , I set up the connection code which connects my Python with SSMS locally (window authentication) .

Afterwards , creating reusable function to export data from python to sql server.

Then I use loop to export my dataframes into ssms in "OlistDB" dataframe.

7. Data Analysis Using SQL

Performing Analysis in SSMS to answer the key business question:

- a) **Total Number of order Completed – 98,816**
- b) **Top 5 State where number of customers are higher -**

	customer_state	No.of customer
1	SP	40302
2	RJ	12384
3	MG	11259
4	RS	5277
5	PR	4882

- c) **Top 5 state with the highest revenue -**

	customer_state	Total_Sales
1	SP	5202955.05
2	RJ	1824092.67
3	MG	1585308.03
4	RS	750304.02
5	PR	683083.76

- d) **Top 5 product Category with the highest revenue -**

	product_category_name	Total_Sales
1	Beleza Saude	1258681.34
2	Relogios Presentes	1205005.68
3	Cama Mesa Banho	1036988.68000001
4	Esporte Lazer	988048.970000007
5	Informatica Acessorios	911954.320000003

- e) **Top 5 product category based on average review rating-**

	product_category_name	avg_review
1	Cds Dvds Musicais	4.64
2	Fashion Roupas Infantil Juvenil	4.5
3	Livros Interesse Geral	4.45
4	Construcao Ferramentas Ferramentas	4.44
5	Flores	4.42

f) Top 5 customer State with the largest delivery days -

	customer_state	avg_delivered_day
1	RR	28.98
2	AP	26.73
3	AM	25.99
4	AL	24.04
5	PA	23.32

g) Percentage of Orders getting late

	delivered_on_time	delivery %
1	Yes	92.42
2	No	7.58

h) Product Category Which experience the higher late delivery percentage -

	product_category_name	late_percent
1	Casa Conforto 2	16.666600
2	Moveis Colchao E Estofado	13.513500
3	Audio	12.154600
4	Artigos De Natal	12.000000
5	Fashion Underwear E Moda Praia	11.811000
6	Casa Conforto	10.023300
7	Eletronicos	8.794400
8	Livros Tecnicos	8.745200
9	Alimentos	8.617200
10	Beleza Saude	8.515500

i) Product Category with revenue above the average category revenue -

	product_category_name	revenue_by_category	avg_revenue
1	Beleza Saude	1258681.34	183670.86
2	Relogios Presentes	1205005.68	183670.86
3	Cama Mesa Banho	1036988.68	183670.86
4	Esporte Lazer	988048.97	183670.86
5	Informatica Acessorios	911954.32	183670.86
6	Moveis Decoracao	729762.49	183670.86
7	Cool Stuff	635290.85	183670.86
8	Utilidades Domesticas	632248.66	183670.86
9	Automotivo	592720.11	183670.86
10	Ferramentas Jardim	485256.46	183670.86
11	Brinquedos	483946.6	183670.86
12	Bebes	411764.89	183670.86
13	Perfumaria	399124.87	183670.86
14	Telefonia	323667.53	183670.86
15	Moveis Escritorio	273960.7	183670.86
16	Papelaria	230943.23	183670.86
17	Pcs	222963.13	183670.86
18	Pet Shop	214315.41	183670.86
19	Instrumentos Musicais	191498.88	183670.86
20	Eletrportateis	190648.58	183670.86

j) Category Contribution in total revenue in percentage -

	product_category_name	revenue_by_category	revenue	contribution_by_category
1	Beleza Saude	1258681.34	13591643.7	9.26
2	Relogios Presentes	1205005.68	13591643.7	8.87
3	Cama Mesa Banho	1036988.68	13591643.7	7.63
4	Esporte Lazer	988048.97	13591643.7	7.27
5	Informatica Acessorios	911954.32	13591643.7	6.71

k) Month – over – month revenue growth percentage -

	purchase_year	purchase_month	revenue_by_month	previous_revenue_by_month	growth_by_month_percent
1	2016	10	49507.6600000002	267.36	18417.2277079594
2	2016	12	10.9	49507.6600000002	-99.9779832050232
3	2017	1	120312.87	10.9	1103687.79816513
4	2017	2	247303.019999995	120312.87	105.549929945148
5	2017	3	374344.300000001	247303.019999995	51.3706949474405
6	2017	4	359927.230000001	374344.300000001	-3.85128610212559
7	2017	5	506071.140000001	359927.230000001	40.6037381500722
8	2017	6	433038.600000005	506071.140000001	-14.4312793651904
9	2017	7	498031.480000001	433038.600000005	15.0085650563263
10	2017	8	573971.680000014	498031.480000001	15.248072270452
11	2017	9	624401.690000015	573971.680000014	8.78614951873579
12	2017	10	664219.430000018	624401.690000015	6.37694302204757
13	2017	11	1010271.370000004	664219.430000018	52.0990390178755
14	2017	12	743914.170000002	1010271.370000004	-26.3649161907862
15	2018	1	950030.360000038	743914.170000002	27.7069853367643
16	2018	2	844178.710000029	950030.360000038	-11.1419228749705
17	2018	3	983213.440000033	844178.710000029	16.4698218935181
18	2018	4	996647.750000033	983213.440000033	1.36636761189917
19	2018	5	996517.680000027	996647.750000033	-0.0130507493752162
20	2018	6	865124.310000021	996517.680000027	-13.185252267677
21	2018	7	895507.220000023	865124.310000021	3.51197043578635
22	2018	8	854686.330000025	895507.220000023	-4.55840992549411
23	2018	9	145	854686.330000025	-99.9830347116936

l) State with the higher cancelation rate of orders -

	Customer_state	percent_of_cancel
1	RR	2.173900
2	RO	1.185700
3	PI	0.808000
4	SP	0.783300
5	RJ	0.669100

m) Most used payment method used by Customers

	payment_type	no_of_use
1	Credit Card	76505
2	Boleto	19784
3	Voucher	3866
4	Debit Card	1528
5	Not Defined	3

n) Customer State with the highest percentage of late order delivery

	customer_state	count_by_state
1	AL	22.921900
2	MA	18.828400
3	SE	15.223800
4	PI	15.126000
5	CE	14.855300

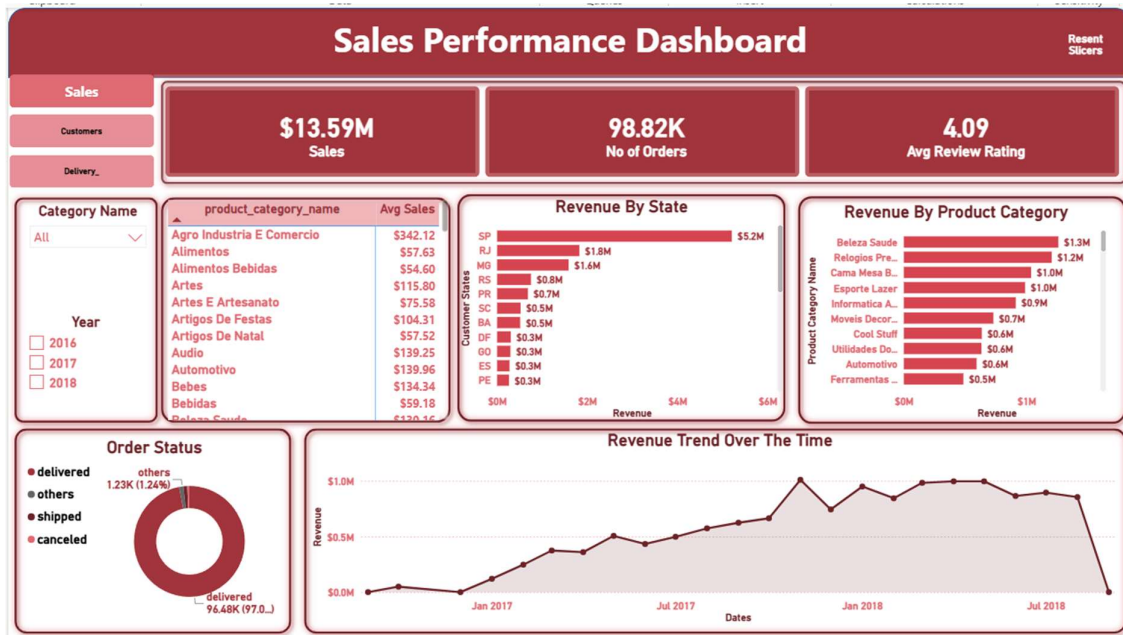
o) Find which combination of customer state and seller state have most percentage of late deliveries -

	customer_state_to_seller	late_delivery_percent
1	AL-AM	100.000000
2	AL-PB	100.000000
3	AC-BA	100.000000
4	AM-CE	100.000000
5	AL-MA	75.000000
6	BA-MS	66.666600
7	RS-RN	60.000000
8	SE-MA	60.000000
9	TO-MA	50.000000
10	CE-MT	50.000000
11	PE-MT	50.000000
12	MT-MS	50.000000
13	AL-PR	46.511600
14	ES-MA	46.153800
15	MA-MA	40.000000
16	TO-DF	40.000000

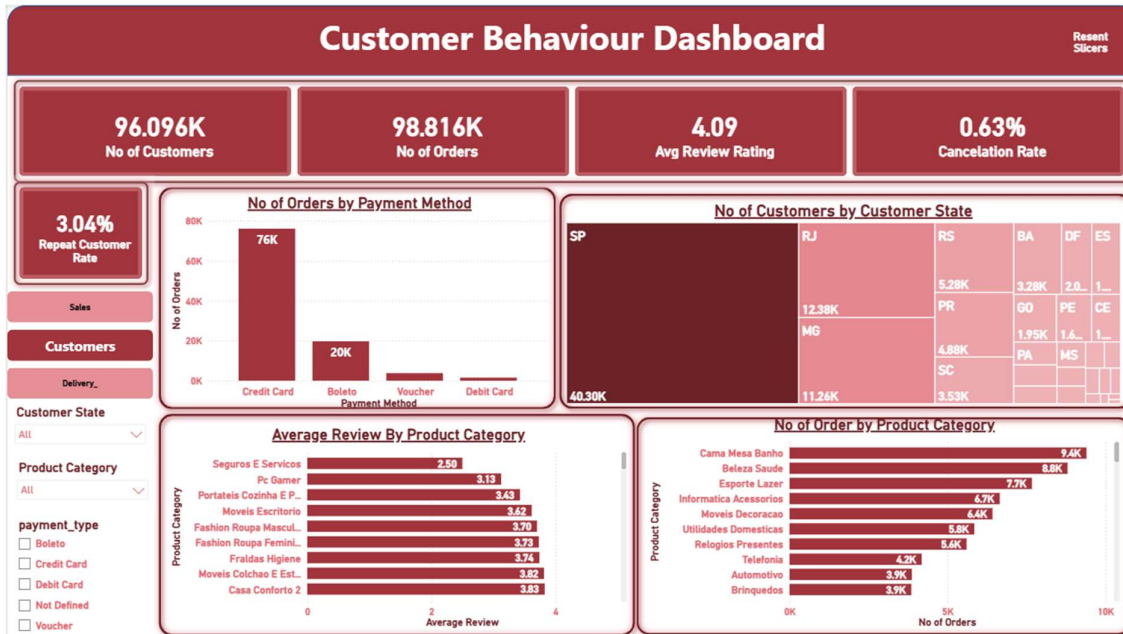
8. Power Bi Dashboards

Finally I build Power Bi interective Dashboard which have 3 pages –

* Sales Performance



* Customer Behaviour



Sales

Customers

Delivery

7.58%

Percentage of Late Delivery

96K

No of Delivered Orders

23.40

Avg Estimated delivered day

Customer State

All

Product Category

All

seller_state

Year

☐ 2016
 ☐ 2017
 ☐ 2018

Percentage of Late Delivery By Seller State

Percentage of Late Delivery By Customer State

Trends of Days Required By Different Stages To Complete An Order

a) **Sales Recommendation**

Majority of Sales is concentrated in only 3 states “SP”, “RJ”, “MG” contribution more the 60% of Total revenue and only these 3 state have a total sales of more then 1 million dollars. So , To increase the sales, the contribution of other states should be increase , which can be possible through marketing.

b) Customer Recommendation

The percentage of repeat customers (customers who place order more than 1 time) is less than 3.05% , which indicates that company have very low percentage of loyal customers or customers can easily shift to other platforms for online purchases,

To increase the repeat customer base, company should come up with **subscription plans**.

the most ordered product by customers are "**Cama Mesa Banho**", "**Beleza saude**" and "**Esporte Lazer**". These product should be use should for **promotion** as most saled and revied product category .

Most of the customers are from "**SP**","**RJ**","**MG**" state , that make sense as these state contribute the most in revenue , this implies the same thing as before , the e-commerce is concentrated only in these three states. **Investment in other states is recommended.**

Most used payment method by customers are credit card. So focus this payment method and make sure that this is available In most of the order payment options

c) Delivery Recommendation

the late delivery percentage is **7.6% approx.**

and customer state which experience the higher percentage of late delivery are

"AL","MA","SE" with average percentage **22.92,18.83,15.22** respectively ,

this indicate that delivery system Is less efficient in these customer states.

on the other hand ,

seller state which experience the higher percentage of late delivery are

"AM","MA","PA".

combination of (customer state – seller state) = (AL-AM),(AL-PB),(AC-BA),(AM-CE) have experience 100% late delivery , so delivery system need to be improve in those states